

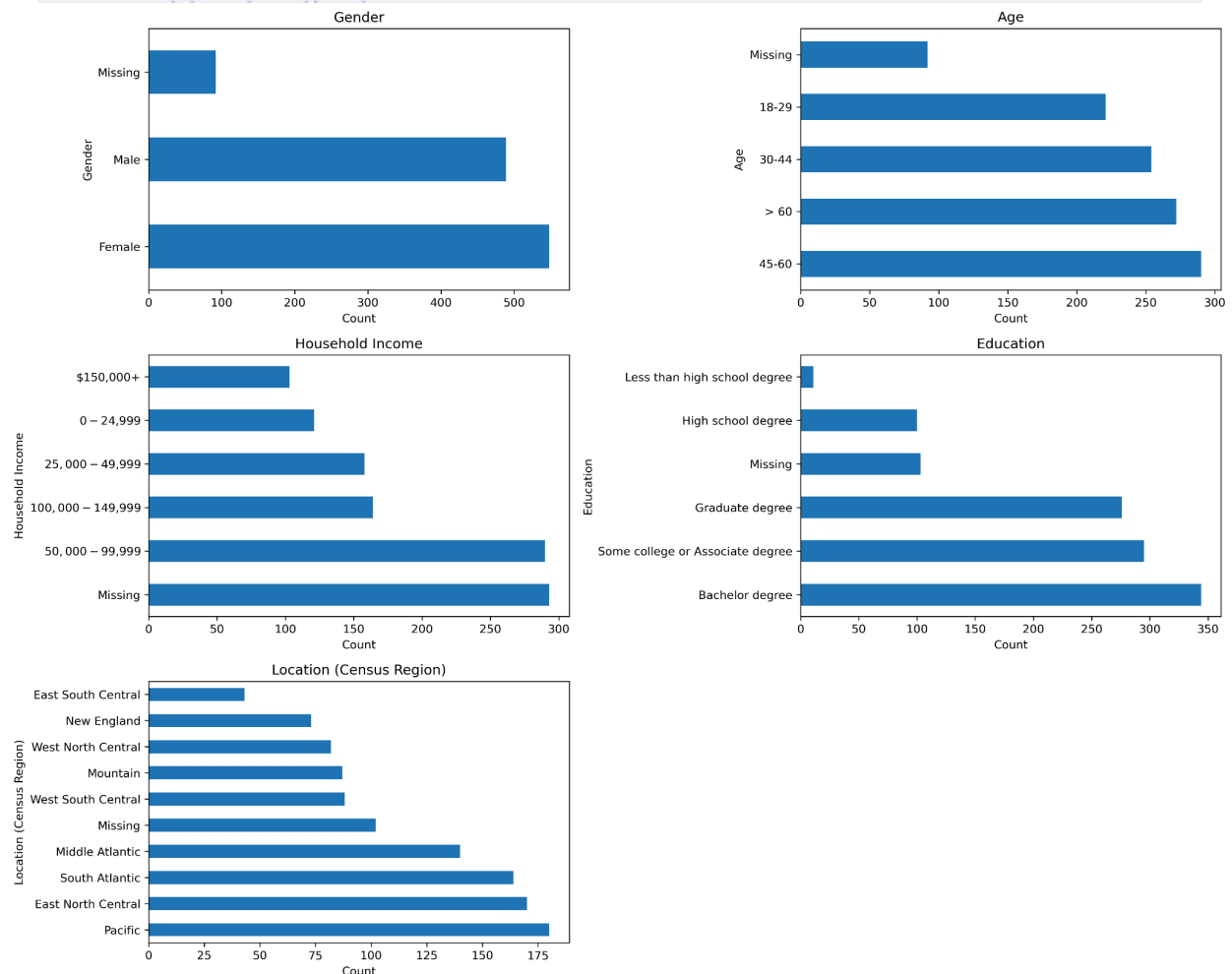
SDS 5380 Social Algorithm Assignment 1 Part 1
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Step 1: Data Exploration

In `survey_analysis.py`, we first explored the overall structures and properties of the dataset. It contains 1129 responses with 13 variables (columns), including 7 substantive survey questions, 5 demographic information about the respondents, and 1 ID column for identification. Most substantive questions have low levels of missing data (generally below 4% with the later questions having an increasing tendency of data missing). Household income has the largest amount of missing data (26%) showing people tend to not share confidential information like household income in a survey. This indicates that missingness is more common for personal demographic information.

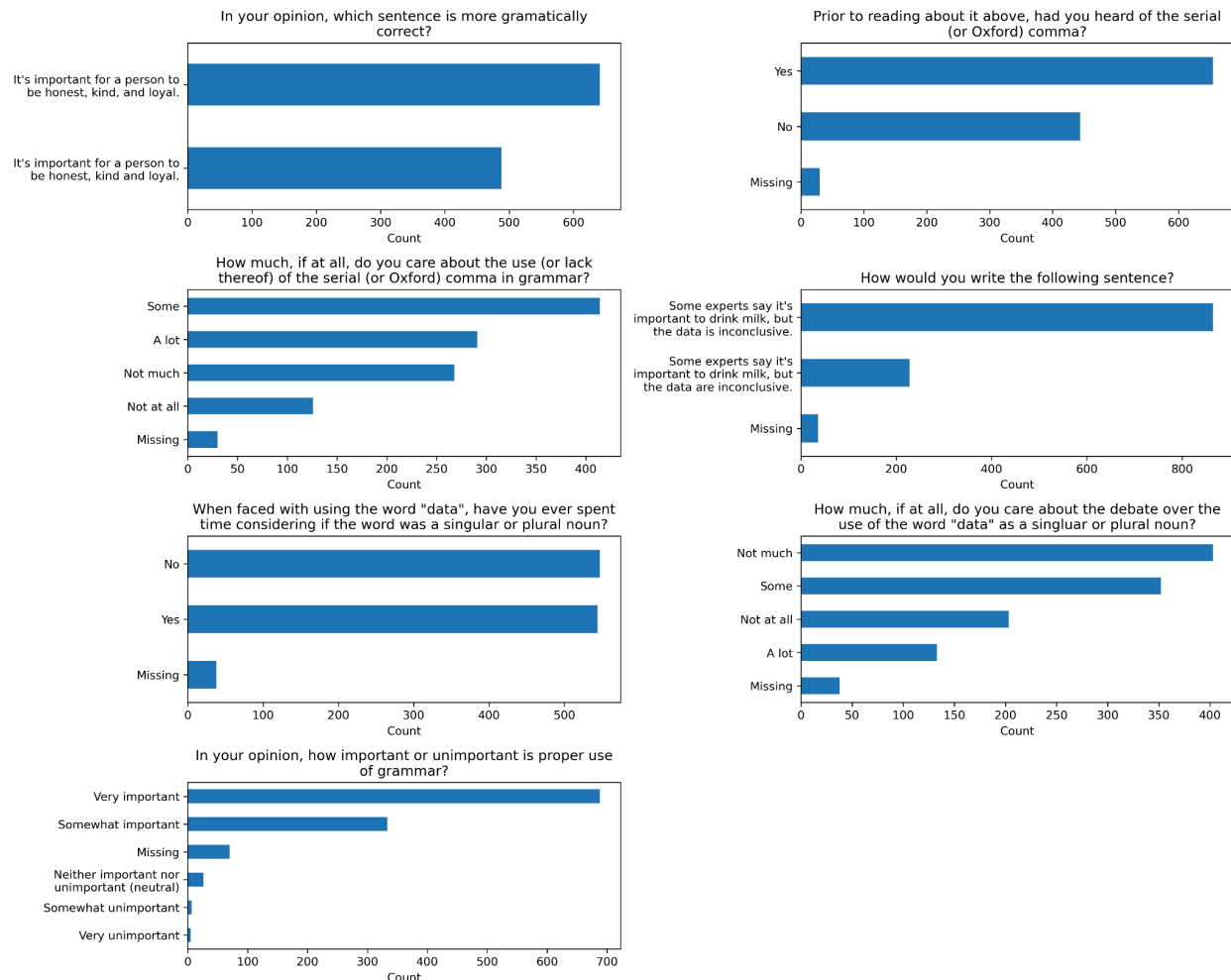
Missing Data Analysis:

	Missing Values	Percentage
RespondentID	0	0.00
In your opinion, which sentence is more gramati...	0	0.00
Prior to reading about it above, had you heard ...	30	2.66
How much, if at all, do you care about the use ...	30	2.66
How would you write the following sentence?	36	3.19
When faced with using the word "data", have you...	38	3.37
How much, if at all, do you care about the deba...	38	3.37
In your opinion, how important or unimportant i...	70	6.20
Gender	92	8.15
Age	92	8.15
Household Income	293	25.95
Education	103	9.12
Location (Census Region)	102	9.03



The age distribution is skewed toward middle-aged and older respondents. The sample is also relatively highly educated, with a large share of respondents reporting at least a bachelor's degree. Household income shows both substantial variation and substantial missingness.

The unadjusted response distributions for the substantive survey questions shows variation in respondents' attitudes toward grammar-related issues. Most respondents prefer the sentence with the Oxford comma and report having heard of it prior to the survey. Views on the use of the word "data" are more mixed, with respondents divided on whether they have considered its singular versus plural usage, although most choose "data is" in the example sentence. Finally, the majority of respondents rate proper use of grammar as important.

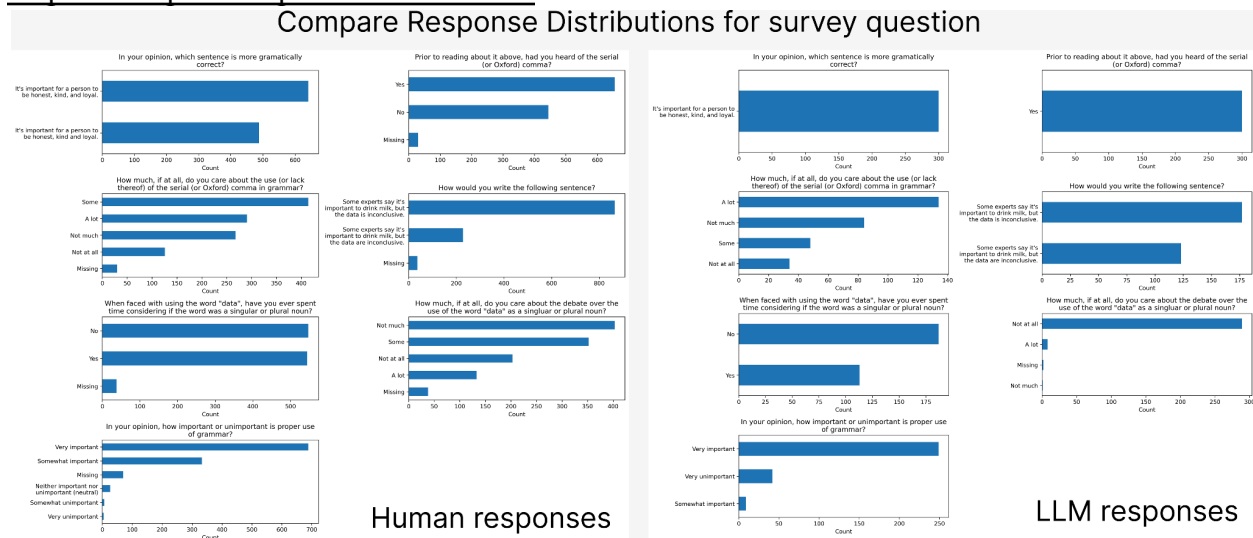


Step 2: LLM Survey responses

We implemented `GPT_prompt.py` using `gpt-4o-mini` to generate survey responses. For each of 300 simulated respondents, we randomly sampled demographic attributes from the baseline survey data and constructed prompts that presented these along with the survey questions. The model was instructed to respond in a multiple-choice format (A-E), which can be parsed and mapped back to the original answer text to create `gpt_comma_survey.csv`.

The original parameters for `client.chat.completions.create` is `n=1` with `temperature = 1.0`. By exploring, it shows that `n` corresponds to the number of responses returned from a single prompt (same "person" with the same demographic information in our case). Setting `n = 1` is appropriate here because we want exactly one response per in silico respondent. The temperature parameter controls the level of randomness in the model's outputs, with higher temperature values leading to greater variation in responses.

Step 3: Compare Response Distributions.



The side-by-side comparison reveals systematic differences between human and LLM responses across all survey questions.

LLMs demonstrate a strong preference for correct answers. For grammatical correctness, LLMs overwhelmingly favor the Oxford comma version, and nearly all claim prior familiarity with the term. These patterns suggest LLMs default to normative grammatical knowledge rather than reflecting realistic population diversity.

Moreover, LLM responses lack the variability present in human data. While humans distribute across multiple response options, showing genuine uncertainty and moderate positions, LLMs concentrate heavily on one or two extreme choices. For example, when asked how much they care about grammar debates, humans spread across "not much," "some," and "a lot," whereas LLMs cluster at "a lot" or "not at all" with minimal middle-ground responses.

This uniformity persists despite providing diverse demographic information to the model, suggesting the LLM does not meaningfully adjust responses based on demographic context. Overall, LLM responses are more certain, more extreme, and less reflective of the natural heterogeneity in human attitudes toward grammar.

Step 4: Post-stratification I on Baseline Data

We used the baseline survey data from `comma-survey.csv`. We trained separate multinomial logistic regression models for each survey question using demographic information (e.g.,

Gender, Age, Income, Education, Location) as predictors. Prior to model training, we dropped rows with missing values in the variables.

Step 5: Post-stratification I on LLM Data

We repeated the same procedure as in Step 4 using the GPT-generated survey data from `gpt_comma_survey.csv`. We trained separate multinomial logistic regression models for each survey question using the same demographic information and preprocessing steps.

Step 6: Census Data Gathering

Following the assignment instructions, we generated the table and copied the corresponding API [TABULATE QUERY URL](#)

To process the Census data, we wrote a Python script that loads the JSON output and applies manual mapping dictionaries to translate encoded Census category values into the human-readable category labels used in our survey. The script converts the API output into a clean tabular format, filters out incomplete rows, and saves the resulting Census population counts as a CSV file. The processed Census counts can then be loaded either by calling the provided function or by reading the saved CSV directly for downstream analysis.

Step 7: Post-stratification II - Baseline and GPT Data

We generated post-stratified estimates using the fitted models from steps 4 and 5. We created 1,800 demographic cells using `itertools.product()` across five demographic information. For each cell, fitted models from steps 4 and 5 predicted response probabilities using `sklearn.linear_model.LogisticRegression.predict_proba` function, which were then weighted by U.S. Census population counts and aggregated. Post-stratified attitudes were saved to `poststratified_attitudes.csv`.

Q1) In your opinion, which sentence is more grammatically correct?					
Response	(i)	(ii)	(iii)	(iv)	(v)
... kind and loyal.	43.237%	—	47.064%	—	—
... kind, and loyal.	56.763%	100.000%	52.936%	—	100.000%
Q2) Prior to reading about it above, had you heard of the serial (or Oxford) comma?					
Response	(i)	(ii)	(iii)	(iv)	(v)
Yes	60.266%	100.000%	52.691%	—	100.000%
No	39.734%	—	47.309%	—	—
Q3) How much, if at all, do you care about the use of the serial comma in grammar?					
Response	(i)	(ii)	(iii)	(iv)	(v)
A lot	26.932%	44.954%	26.157%	36.152%	31.000%
Some	37.560%	16.514%	36.309%	15.802%	20.667%
Not much	25.121%	29.358%	25.919%	35.801%	33.667%
Not at all	10.386%	9.174%	11.614%	12.245%	14.667%
Q4: How would you write the following sentence?					
Response	(i)	(ii)	(iii)	(iv)	(v)
Some ... data is ...	80.121%	60.550%	82.262%	65.852%	72.000%

Some ... data are ...	19.879%	39.450%	17.738%	34.148%	28.000%
Q5: When faced with using the word "data", have you ever spent time considering if the word was a singular or plural noun?					
Response	(i)	(ii)	(iii)	(iv)	(v)
Yes	50.604%	36.697%	46.010%	31.032%	27.667%
No	49.396%	63.303%	53.990%	68.968%	72.333%
Q6: How much, if at all, do you care about the debate over the use of the word "data" as a singular or plural noun?					
Response	(i)	(ii)	(iii)	(iv)	(v)
A lot	11.111%	3.241%	11.337%	2.783%	1.000%
Some	32.005%	—	29.648%	—	—
Not much	38.889%	—	38.104%	—	0.667%
Not at all	17.995%	96.759%	20.911%	97.217%	98.333%
Q7: In your opinion, how important or unimportant is proper use of grammar?					
Response	(i)	(ii)	(iii)	(iv)	(v)
Very important	64.372%	81.193%	64.887%	78.252%	66.667%
Somewhat important	32.246%	3.211%	30.566%	2.886%	2.000%
Neutral	2.536%	—	3.652%	—	—
Somewhat unimportant	0.604%	—	0.669%	—	—
Very unimportant	0.242%	15.596%	0.225%	18.862%	31.333%

Table 1. Pre- and Post-Stratification Estimates for Baseline and GPT Survey Data
(i)-(v) denote, respectively, the raw human survey sample means, GPT-generated sample means under the initial survey demographics, post-stratified estimates for the human population, post-stratified estimates for the GPT population, and GPT-generated sample means using the census population (Step 8)

The baseline survey's final population-level estimates suggest that estimates computed directly from the original data (Table 1. (i)) tend to show higher levels of grammatical knowledge than the post-stratified estimates (Table 1. (iii)). For example, the difference between the sample mean and the post-stratified estimate in question 2 is about 7.575%p. This difference is due to the original data containing a high portion of highly educated respondents (some college education or higher) compared to the U.S. population. While questions related to how much respondents care about grammar show relatively small differences, post-stratification shifts estimates across all questions toward lower levels of grammatical knowledge and interest. Post-stratification adjusts the overstatement of grammatical knowledge and interest present in the original survey estimates.

If we examine the GPT-generated data estimates (Table 1. (ii) and (iv)), post-stratified estimates cannot be obtained for question 1 and 2 because the GPT generated a single option for all respondents. Additionally, for question 6 and 7, some response options are absent and responses are highly concentrated on a single option, making meaningful comparisons between estimates difficult. Although post-stratification shifts the estimates toward lower levels of grammatical knowledge and interest, similar to the baseline data results, these changes are difficult to interpret

as meaningful adjustments due to the inherent problems in the GPT data discussed in Step 3 (e.g., lack of response variability, tendency toward normatively correct answers).

Step 8: Validation Survey for Census Population

We reused the code structure from Step 2, but replaced the uniform sampling with weighted sampling based on census population counts from Step 6. Specifically, we used `numpy.random.choice()` with probabilities proportional to each demographic cell's population count. We generated 300 responses and saved them as `gpt_census_survey.csv`.

Step 9: Observations

Table 1 in Step 7 summarizes the results of raw and post-stratified estimates for baseline survey, GPT-generated data using the initial survey demographics, and GPT-generated data using the census population, which are compared in Step 9.

The comparison between Table 1 column (i) and (iii) is the comparison of pre- and post-stratification for the human survey data. As discussed in Step 7, post-stratification (Table 1. (iii)) adjusts the overstatement of grammatical knowledge and interest present in the raw human survey estimates (Table 1. (i)). To be specific, the raw human survey's sample means shows higher levels of grammatical knowledge than the post-stratified estimates for the human population. According to Step 1, this difference is due to the overrepresentation of respondents with some college education or higher in the raw human survey sample compared to the U.S. population.

Comparing raw and post-stratified estimates for GPT-generated data (Table 1. (ii), (iv), and (v)) suggests that inherent limitations of the GPT-generated data cannot be resolved by post-stratification or by changing the demographic information used to generate the data. If we explore estimates for GPT-generated data, sample means and post-stratified estimates for question 1 and 2 are absent because the GPT generated a single option for all respondents. Moreover, estimates of some response options in question 6 and 7 are missing. This is because GPT-generated responses are more extreme, less variable, and biased toward normatively correct answers as we discussed in Step 3.

Furthermore, although post-stratification (Table 1. (iv)) and GPT-generated data using the census population (Table 1. (v)) both shift the raw GPT estimates toward lower grammatical knowledge and interest, all GPT-generated data estimates (Table 1. (ii), (iv), and (v)) seem similar to each other compared to human survey estimates (Table 1. (i) and (iii)). This similarity indicates that the LLM cannot internally reflect the natural heterogeneity of human attitudes, even when the settings are changed.

SDS 5380 Social Algorithm Assignment 1 Part 2

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Question: What cultural knowledge is "common sense" to native-born Americans but must be explicitly learned by immigrants? Can search query patterns reveal the scale and priority of these hidden knowledge gaps?

Dataset of choice: web search queries

Background and Motivation

Cultural adaptation always involves two types of knowledge: one is the explicit rules that have been formally taught and conveyed, and the other is the tacit knowledge absorbed by local residents through long-term immersion. Although these explicit rules have detailed records and are taught through schools, communities, or government services, this tacit knowledge is largely unseen as immigrants need to figure things out on their own without guidance. Utilizing web search data makes it possible to identify forms of tacit knowledge that immigrants face in their daily lives. Moreover, understanding the scale and urgency of these hidden knowledge gaps can inform proper actions to support immigrants more effectively.

Methodology & Analysis:

The large-scale search query logs will be analyzed using content-based filtering to identify groups that may include immigrants or international students. This filtering will rely on indicators such as multilingual queries, immigration related keywords, and searches related to newcomer experiences (e.g, "How to tip" or "What is a potluck"). These queries will then be sorted based on search frequency to estimate the scale and priority of different knowledge gaps, and then identify the systematic mismatch between institutional emphasis and immigrants' actual challenges. This comparison and analysis lead directly to practical suggestions.

Benefits & Downsides over traditional data sources:

In comparison to traditional data sources like surveys and interviews, online search queries have several advantages. Firstly, search queries enable the use of large-scale data at low cost compared to traditional data sources. Moreover, search queries can provide natural and authentic information by automatically capturing real-time needs without the intervention of researchers. Thus, they reveal the problems that people indeed encounter in their daily lives while preventing the embarrassment often present in traditional settings. In addition, search frequency and query refinement patterns provide an implicit priority measurement standard: issues that generate a large number of searches are likely to represent widespread and urgent challenges.

However, there are also some limitations to the search data. It overly reflects the groups of people who are younger, more dependent on digital technology, and more accustomed to using search engines. Moreover, it cannot directly indicate whether individuals have successfully resolved their confusion after searching. Therefore, these findings should be interpreted as evidence of knowledge gaps, rather than proof of behavioral change or successful integration. Despite these limitations, web search queries have enabled the large-scale and real-time exposure of hidden cultural knowledge gaps, providing a unique and powerful supplement to traditional methods.